

L9: SIS Model – No Epidemic Threshold For Scale-Free Nets

Let us now examine the epidemic threshold for two-degree distributions we have studied considerably in the past.

1) For random networks with Poisson degree distribution (such as ER networks), the variance is equal to the mean, and so the second moment is: $\bar{k}^2 = \bar{k}(1 + \bar{k})$

So, we have an epidemic if $\frac{\beta}{\beta + \mu} > \frac{1}{\bar{k} + 1}$, which is equivalent to the expression we derived under homogeneous mixing ($\frac{\beta}{\mu} > \frac{1}{\bar{k}}$).

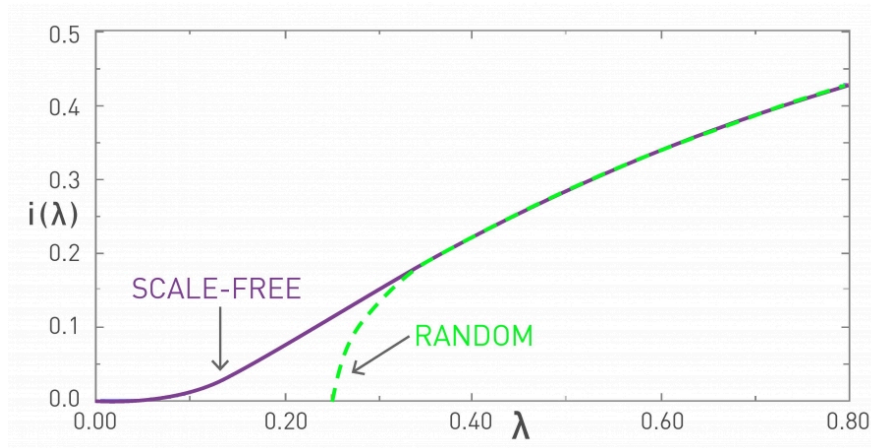


Figure 10.11 from *Network Science* by Albert-László Barabási.

In the visualization, the x-axis parameter λ refers to the ratio $\frac{\beta}{\beta + \mu}$. In the "**random network**" curve (green), if that ratio is larger than $1/(1 + \bar{k})$ an outbreak will lead to an epidemic. The y-axis value shows the steady-state fraction of infected individuals in the endemic state.

It is important to note that if λ is less than the threshold $\frac{1}{1 + \bar{k}}$, then the outbreak will die out and it will not cause an epidemic.

2) For networks with a power-law degree distribution ("scale-free network" curve shown in purple), and with an exponent γ between 2 and 3, the variance (and the second moment) of the degree diverges to infinity ($\bar{k}^2 \rightarrow \infty$).

This means that the condition for the outbreak of an epidemic becomes:

$$\lambda > \frac{\bar{k}}{\bar{k}^2} \rightarrow 0$$

This is a remarkable result with deep and practical implications. It states that if the contact network has a power-law degree distribution with diverging variance, then **any outbreak will always lead to an epidemic**, independent of how small λ is. Even a very weak pathogen, with a very small λ , will still cause an epidemic.

The fraction of infected individuals in the endemic state still depends on this ratio – but whether we will get an endemic state or not does **not** depend on λ .

The reason behind this negative result is the presence of hubs – nodes with a very large degree. Such nodes get infected very early in the outbreak – and then they infect a large number of other susceptible individuals.

Food For Thought

Suppose that the ratio λ is equal to 1/4. Plot the fraction of infected individuals of degree k in the endemic state as the ratio $\frac{\bar{k}}{\bar{k}^2}$ varies between 0 and 1/4.